



University
of Glasgow

TODO DATE AND TIME

Duration: H-Level: 1.5 hours, M-Level: 2 hours

Additional time: 30 minutes

Timed exam – fixed start time

DEGREE of MSc

Big Data: Systems, Prog & Mgt

COMPSCI 4064 (H) / 5088 (M)

(H-Level: Answer the first 3 Questions, M-Level: Answer all 4 Questions)

This examination paper is an open book, online assessment and is worth a total of 75 marks.

1. (a) **Context:** Consider the following scenario. You are working at the British Broadcasting Company (BBC) and are responsible for preparing on-demand video recordings from live programming. An hour of broadcast material uses around 45GB of storage space, and the BBC runs at peak times 35 concurrent channels. Audio and video channels are stored separately, and around half of the recorded content have subtitles in multiple languages. You need to take the high-quality recordings, encode and downsample them to 6 resolution and bit-rate options. The BBC has a target of having on-demand videos published to their iPlayer offering within 60 minutes of the live recording ending.

Question: The preparation of these videos constitutes a big data task. List the dimensions/properties of the task that makes it a big data problem and discuss why.

Guidance: The question is worth 6 marks. There are 3 marks for correctly identifying the dimensions/properties and 3 marks for the associated discussion (1 mark per correct point made).

[6]

Solution:

- Volume [1 Mark]: Up-to 1,575GB of video per hour constitutes a large volume of data to be processed and each video needs to be converted into 6 different forms [1 Mark]
- Variety [1 Mark]: You have multiple types of data - video, audio and text, which need to be combined into the final output [1 Mark].
- Variability [1 Mark]: The amount of data needing processed varies over time based on the number of channels that are currently broadcasting [1 Mark].

- (b) **Context:** When building a solution to this problem, you might tackle it using either batch or stream processing.

Question: Discuss which is a better solution and why.

Guidance: The question is worth 4 marks. You must state what type of solution you are arguing for, and then provide 2 distinct points for why this solution is superior to the alternative, 2 marks per point. For each point you need to contrast against the alternative to receive the marks.

[4]

Solution:

If arguing for batch processing valid points include:

- Batch based processing will be less prone to error [1 Mark], as all of the video, audio and subtitle data will be complete, rather than being streamed from the recording set. [1 Mark]
- Batch based processing may be more parallelizable (depending on codec used) [1 Mark]: as different segments of a recording can be processed concurrently [1 Mark].

If arguing for stream processing valid points include:

- Stream processing may provide lower latency [1 Mark], as the encoding can happen concurrently with the live show, rather than having to wait for the show to end before encoding. [1 Mark]
- Stream processing (potentially) avoids having to store the high-quality recordings [1 Mark], as the encodings are produced live, saving drive space. [1 Mark].

- (c) **Context:** The BBC decides to develop a horizontally scalable batch processing solution using Hadoop MapReduce. The design for the solution is as follows. For each live video recording, that video will be split into batches of image frames to be processed, where each batch ends in a key frame. The batch of frames is sent to an encoder, which downsamples each frame to a lower resolution. Once all frames in a batch have been processed, they are combined together to create a key-frame block, which also involves crops out (removes) static content between frames to enable compression. The encoded video is a sequence of these keyframe blocks.

Question: Describe using MapReduce terminology how a single video would be processed.

Guidance: The question is worth 14 marks. You are expected to describe 7 steps. There are 7 marks for correctly identifying the steps, and 7 marks for describing what happens within each step.

[14]

Solution:

- **Read 1:** The live video recording will be read from the HDFS as a series of HDFS blocks [1 Mark], where each block represents a batch of image frames to be processed. [1 Mark]
- **Map:** The first (and only) Map task [1 Mark] executes the encoder per image frame, which downsamples each frame to a lower resolution [1 Mark].
- **Sort:** The encoded images are sorted by the batch that they came from [1 Mark] using the key-frame identifier as a key [1 Mark].
- **Write 1:** The ordered image frames are written down [1 Mark] to the local HDFS data node [1 Mark].
- **Read 2:** The ordered image frames are read in by key [1 Mark], the blocks are copied to the machine doing the reduce and recursively merged [1 Mark].
- **Reduce:** The first (and only) Reduce task [1 Mark] reads all of the frames for a key-frame and builds the key-frame block [1 Mark].
- **Write 2:** The key frame blocks are written to the local HDFS data nodes [1 Mark] as output of the reducer (which will subsequently get replicated) [1 Mark].

2. (a) **Context:** A start-up company is designing a new low-cost cloud compute platform using Apache Spark. The core idea of the start-up is to allow members of the public to sell compute time on their mobile phones when those devices are idle, where the compute capacity of multiple phones can be bundled and sold on to a business for profit. The start-up company runs a series of Spark Masters on their own servers hosted in their headquarters, where the Spark Workers are deployed on thousands of mobile phones spread out over the UK.

Question: Discuss why the company might have chosen to use Apache Spark rather than Hadoop as its big data processing framework.

Guidance: This question is worth 8 marks. 4 points are expected, where one mark is available for providing a relevant difference between Spark and Hadoop and 1 mark is available for why this is important in the context of this application.

[8]

Solution: We expect the students to describe the main advantages that Spark has over Hadoop, but avoiding any that are not relevant for this application. Valid points include:

- Hadoop is tied to the HDFS for data storage [1 Mark], which would not be effective if deployed on mobile phones as mobile data upload costs are high [1 Mark].
- Spark does not write intermediate data to disk [1 Mark], so those costs will be avoided [1 Mark].
- Tasks in Spark are generally more efficient in Hadoop (up-to 100x faster) [1 Mark], which means less local compute (and hence battery drain) [1 Mark].
- Spark supports more programming languages [1 Mark], and as such has a wider customer base [1 Mark].

A point discussing more efficient scheduling or failure recovery would also be valid answers here.

- (b) **Question:** The company runs a series of tests and discovers that often tasks are getting executed multiple times, which is wasting resources. Given your understanding of how Spark executes jobs, what reasons are there that might cause this to happen?

Guidance: The question is worth 9 marks. We expect you to identify three different possible causes. For each, there is 1 mark for describing a valid cause, 1 mark for specifying where in the Spark job execution this might happen, and 1 mark for discussing why this is a particular problem for the company's scenario.

[9]

Solution: Valid Answers include:

- The mobile phone runs out of memory [1 Mark], which will happen during task execution [1 Mark] because mobile phones typically have only a small amount of RAM that may also be shared with other applications [1 Mark].
- The task scheduler fails to receive heartbeats from the executor [1 Mark] due to

the phone losing network access [1 Mark], which is likely to occur given poor signal in some areas.

- The phone running the executor shuts down [1 Mark] due the battery running out [1 Mark], which is likely given that phones are not designed for heavy-weight processing over extended periods.
- The phone carrier stops the data download or upload [1 Mark] due to data limits on the contract [1 Mark], since unlike in a normal deployment setting network IO is strictly limited.

(c) **Context:** An engineer at the company suggests that moving to an Apache YARN based management system might reduce the need for services managed at the headquarters.

Question: Is this true and explain why. Then provide a short discussion on how this would affect the robustness of the companies solution.

Guidance: This question is worth 6 marks. We expect you to specify whether the statement is true [1 Mark], and then summarize why [2 Marks]. You should then include a paragraph describing how using YARN changes how applications are managed using Spark [2 Marks], and how this affects robustness in this scenario [1 Mark].

[6]

Solution:

The statement is true [1 Mark], as using YARN Spark jobs are managed via an Application Master [1 Mark] that is deployed onto the worker nodes/node managers (on the phones in this case) [1 Mark]. The company will still need a resource manager running at the headquarters [1 Mark], but when a business submits a job, that will trigger the creation of an Application Master on a Node Manager that will manage the job from then on [1 Mark]. Given the potential instability and number of failure modes for a mobile phone (e.g. from power or network loss), this will likely make the company's solution less robust [1 Mark].

3. (a) **Context:** The School of Computing Science is preparing to purchase a new compute cluster and data storage solution to support undergraduate and PhD students. The system is expected to support 1000 undergraduate students and 100 PhD students. A typical undergraduate student is expected to require 50GB of file storage, while a PhD student is expected to need 1TB of file storage. Undergraduate students rarely access their data, but when they do, read/write latency is important. Meanwhile, PhD students typically are running experiments on large data, which requires high read throughput of large blocks of data.

Question: The School is considering buying a RAID1 solution. Why might the School want a RAID6 solution over a RAID1 solution?

Guidance: This question is worth 4 marks. 2 marks are available for describing the differences between RAID1 and RAID6 and 2 marks are available for valid reasons for why RAID6 is superior for this scenario.

[4]

Solution:

The relevant differences between RAID1 and RAID6 are:

- RAID1 simply creates a second mirrored copy of the provided data, while RAID6 uses striping to distribute segments over multiple drives [1 Mark].
- RAID1 is secure against 1 disk failure, RAID6 (depending on configuration) is secure against multiple disk failures. [1 Mark].
- RAID1 is less space efficient than RAID6 when achieving redundancy, as RAID6 uses parity data rather than direct replication [1 Mark].

The reasons that RAID 6 is better for this scenario are:

- Given the large number of students and storage needed, RAID1 would be very costly to deploy [1 Mark].
- RAID6 due to striping can enable parallel reads of the data via large numbers of disks for the PhD students. [1 Mark].
- RAID6 has better security from multiple disk failures [1 Mark].

- (b) **Question:** A member of staff suggested that instead of using RAID-based storage they should instead use an HBase installation. Is this a good idea given the requirements?

Guidance: This question is worth 4 marks. You are expected to make two points in your answer, marks are awarded for highlighting the difference between the two solutions and why this is important given the requirements.

[4]

Solution: This is a logic check to catch students entering queries into answer generators. The correct answer is:

- HBase is a column store not a file store [2 Marks] and so cannot be used to store the student's files [2 Marks].

Answer generators will produce a bunch of irrelevant information regarding Complexity, Performance, Cost and Learning curve. Failure on check questions are used to flag exam scripts for deeper analysis.

- (c) **Context:** The School buys 50 compute nodes with 24 CPU cores per node, each with an internal 1TB SSD for use with a Cassandra installation that was requested by one of the research groups.

Question: When configuring Cassandra should the School use virtual nodes or not?

Guidance: This question is worth 5 marks. 2 marks are for describing what virtual nodes do, 1 mark is for correctly identifying whether this is needed and 2 marks are for explaining why.

[5]

Solution:

Virtual nodes allow for one physical node to be divided into multiple nodes for the purposes of selecting which nodes to store data on [1 Mark]. They are primarily used to enable better load balancing [1 Mark]. With 50 nodes, it is probably not needed [1 Mark], since 50 unique locations should be highly fault tolerant [1 Mark]. However, if the hashing scheme used co-locates too much data into a subset of the nodes, then virtual nodes may still be needed [1 Mark].

4. (a) **Context:** The typical replication factor in HDFS is 3, meaning that each block will be stored on at least 3 nodes. Keeping in mind the various protocols employed by HDFS (read, write, replication, recovery, etc.):

Question: What would you expect the effects of a higher replication factor to be on *write throughput*?

Guidance: This question is worth 5 marks, of which 2 marks are for identifying the key aspect of the HDFS that affects write throughput, 2 marks are for describing how changing the replication factor affects this and 1 mark is for stating what effect a higher than 3 replication factor would have.

[5]

Solution:

HDFS creates replicas in a pipelined fashion [2 marks], meaning that N replicas can be created with an overhead of $\sim N * \text{network latency}$ over the time it takes to create the first replica (which, for N=3 and a datacentre utilising $\geq 1\text{Gbps}$ links, is negligible) [2 mark]. As such, write throughput would be only slightly reduced. [1 mark]

- (b) **Question:** What would you expect the effects of a higher replication factor to be on *read latency*?

Guidance: This question is worth 4 marks, of which 2 marks are for identifying the key aspect of the HDFS that affects read latency and 2 marks are for describing how changing the replication factor affects this.

[4]

Solution: HDFS reads data from the replica that is the closest (in network distance) to the requesting client [2 Marks]. As such, a higher replication factor would result in possible reductions in read latency [2 marks].

- (c) **Question:** How would you expect a higher replication factor to *affect the NameNode and DataNodes in terms of resource usage*?

Guidance: This question is worth 6 marks. Three points are expected, two marks per point.

[6]

Solution: The only real downside of a higher replication factor would be increased disk usage across the DN's [1 mark], as obviously more copies of the same data will be possibly stored (up to a maximum of number of DN's) [1 mark], and higher memory usage at the NN (as the in-memory fs image will now need to also store the locations of the additional replicas) [2 mark]. Also, more bandwidth will be used during the sending of blockreports from the DN's to the NN's. [2 mark]